Ledger Celo Application

Ledger Application Review

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Contents

1 Project Information . . . **3**
1.1 Document history . . . 3
1.2 Contacts . . . 3

2 Introduction . . . **4**
2.1 Context . . . 4
2.2 Objectives . . . 4
2.3 Scope . . . 4
2.4 Methodology . . . 5
2.5 Version information . . . 5

3 Configuration and compilation . . . **6**
3.1 Application flags . . . 6
3.2 Compilation warnings . . . 7
3.3 Tests . . . 8
3.4 GitHub Actions . . . 12

4 Static analysis . . . **13**
4.1 Clang static analyzer . . . 13
4.2 CodeQL analysis . . . 13

5 Manual review . . . **14**
5.1 APDU overview . . . 14
5.2 Transaction fields displayed . . . 14
5.3 Code review . . . 15

6 Fuzzing . . . **21**

7 Conclusion . . . **22**
7.1 Issues summary . . . 22
7.2 Recommendations . . . 22

8 Ledger summary . . . **24**
8.1 Purpose of the update . . . 24
8.2 Issue summary . . . 24
8.3 Version information . . . 24
8.4 Ledger DevPortal checklist . . . 25
8.5 Production Ready . . . 25
8.6 Remarks . . . 25

Appendix . . . **26**

1 Project Information

1.1 Document history

Version	Date	Details	Authors
1.0	2025-10-24	Initial audit	Elouan Wauquier

1.2 Contacts

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2 Introduction

2.1 Context

Cyriliance Labs Inc. requested Quarkslab to conduct an audit of its "Celo Application" in the context of Ledger's approval framework[1]. This document is the audit report, delivered in the context of Quarkslab's technical proposal[2] and for Ledger approval purpose.

In the process of a Ledger embedded app submission, Cyriliance Labs Inc. wants to assess the security of its "Celo Application" for the Ledger hardware wallets before a public deployment on the Ledger platform. Any application or plugins that are going to be published on the Ledger store must be strictly verified from a security perspective. Quarkslab has been requested by Cyriliance Labs Inc. to execute this security assessment according to Ledger's security guidelines and rules.

2.2 Objectives

The security review is linked to various technical objectives as defined by Ledger:

- Basic security compliance checks (compilation flags, permissions…).
- Detecting potential threats in the code base by static analysis (manual or automated).
- Implementing a fuzzing process.
- Reporting flaws, vulnerabilities, and any other information to maintainers.

The execution of the assessment is fully handled by Quarkslab according to Ledger's security guidelines for embedded application release on Ledger Live[3]. Quarkslab's 3rd-party intervention is neutral regarding security validation, and Ledger owns the unique role of deciding if an application is compliant to their security rules or not, based on Quarkslab's assessment results. Quarkslab shall bear no responsibility either in Ledger's decision, nor in the relevance of the executed assessment with respect to the global security assessment of the targeted application.

2.3 Scope

The app reviewed is "Celo Application". The evaluation started on 2025-10-14 and ended on 2025-10-24. It was performed by one engineer, and managed by a project manager, leading relationship with the customer and Ledger.

[1] Quarkslab's offer, reference 24-06-1674-PRO.
[2] Ledger Externalized security audit framework, October 2022.
[3] https://developers.ledger.com/docs/device-app/deliver/deliverables/security-audit

2.4 Methodology

Category	Action
Application privileges	• Check application flags. • Check allowed derivation paths.
Compilation	• Check compilation warnings. • Check for silenced warnings.
Testing	• Test suite check (no errors). • Test suite validation / coverage analysis.
Static analysis	• Check for defects using `scan-build` and Ledger scan options. • Usage of CodeQL: check with the "security and quality" queries. • Integration in CI if not present.
Manual code review	• List every transaction field. Look which ones must be displayed to the user. • Check transaction parser, field formatters and if sensitive data is properly erased. • Blind signing is not allowed.
Fuzzing	• Implement a transaction fuzzer. Best effort to reach decent coverage. • Use libFuzzer if possible.
Reporting	• Report and executive summary detailing findings, even tests that gave no results. • Feedbacks on the SDK to improve overall security.

2.5 Version information

Initial audit (1.0)	
Repository	https://github.com/blooo-io/app-celo-spender.git
Version	`N/A`
Commit	`55fdbcaad1336d121a3cf8895c0d3ee5ba933a1f`

3 Configuration and compilation

3.1 Application flags

Application flags[4] are set in the `Makefile`, and automatically by the Ledger SDK's `Makefile.devices` using notably `ledger_app.toml`. Since swap operations are not supported, `HAVE_APPLICATION_FLAG_LIBRARY` (`0x0800`) is not set.

Device	DERIVE_MASTER	GLOBAL_PIN	BOLOS_SETTINGS	LIBRARY	NOT_REVIEWED
Nano S Plus	✗	✗	✗	✗	✗
Nano X	✗	✗	✓	✗	✗
Stax	✗	✗	✓	✗	✗
Flex	✗	✗	✓	✗	✗

The `HAVE_APPLICATION_FLAG_BOLOS_SETTINGS` flag lets the application read and modify system parameters such as the device's name. This flag is commonly set to support BLE.

The curve is set to `secp256k1`.

The derivation paths are set to `m/44'/52752'`, `m/44'/60'/0'/0/0`, `m/44'/60'/0'`, and `m/44'/60'/0'/0`. Note that SLIP-0044 assigns `m/44'/52752'` to Celo (CELO), and `m/44'/60'` to Ethereum (ETH).

The parameters set for `CELO` in LedgerHQ's application database do NOT correspond to the flags and parameters set in the application.

INFO	**QB-1** Outdated flag in LedgerHQ's application database.
Perimeter	Configuration

Description

The LedgerHQ's application database expects the `HAVE_APPLICATION_FLAG_LIBRARY` (`0x0200`) to be set for all targets, but it was removed in `c5f913a`[5].

Recommendation

Ledger should update the database entry corresponding to CELO.

Relevant build instructions are provided in Appendix A.

[4] https://developers.ledger.com/docs/device-app/references/app-permissions#application-flags
[5] Commit hash `c5f913a3de3529117aa0f4b7e4c4126a97dbdbb8` on 2025-06-23 for version 1.4.0.

3.2 Compilation warnings

The application is written only in C.

INFO	QB-2 C compiler raises 1 warning.
Perimeter	Compilation

Description

Clang raises 4 warnings, with no security impact.

```
$ docker run --pull always --rm -t -v .:/app \
    ghcr.io/ledgerhq/ledger-app-builder/ledger-app-builder-lite \
    make BOLOS_SDK=/opt/stax-secure-sdk
latest: Pulling from ledgerhq/ledger-app-builder/ledger-app-builder-lite
Digest: sha256:feb1f6bb20a79b54b7e55679e68ab492509328591910ab2000b3b57c48dc7f46
Status: Image is up to date for ghcr.io/ledgerhq/ledger-app-builder/ledger-app-builder-lite:latest
Prepare directories
[GLYPH] Compiling...
[CC]   build/stax/gen_src/glyphs.o
[CC]   build/stax/obj/app/src/apdu/dispatcher.o
/app/src/apdu/dispatcher.c:19:14: warning: unused variable 'sw' [-Wunused-variable]
    uint16_t sw;
             ^
1 warning generated.
  ✂
[CC]   build/stax/obj/app/src_features/signMessageEIP712/commands_712.o
/app/src_features/signMessageEIP712/commands_712.c:140:46: warning:
unused parameter 'flags' [-Wunused-parameter]
                                  uint32_t *flags) {
                                            ^
/app/src_features/signMessageEIP712/commands_712.c:210:44: warning:
unused parameter 'flags' [-Wunused-parameter]
                                 uint32_t *flags) {
                                           ^
/app/src_features/signMessageEIP712/commands_712.c:281:77: warning:
unused parameter 'flags' [-Wunused-parameter]
uint16_t handle_eip712_sign(const uint8_t *cdata, uint8_t length, uint32_t *flags) {
                                                                            ^
3 warnings generated.
  ✂
[CC]   build/stax/obj/app/src_features/signMessageEIP712_v0/cmd_signMessage712.o
/app/src_features/signMessageEIP712_v0/cmd_signMessage712.c:18:51: warning:
unused parameter 'flags' [-Wunused-parameter]
                                  unsigned int *flags) {
                                               ^
1 warning generated.
```

Recommendation

Remove the unused variable and unused parameters, or mark them with `__attribute__((unused))` if this is intended.

3.3 Tests

The codebase includes C unit tests, and Ragger-based integration tests. Ragger is an integrationg testing framework for Ledger devices developped in Python. It tests the typical application flow against a set of snapshots.

3.3.1 Unit tests

Unit tests can be run using CMake and CTest. 2 unit tests are defined in a `CMakeLists.txt`:

- `[FAIL] test_uint256`
- `[PASS] test_tx_parser`

INFO	QB-3 Unit test `test_uint256` fails to compile.
Perimeter	Compilation, Test

Description

```
../tests/unit $ cmake -B build
-- The C compiler identification is GNU 12.2.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Configuring done
-- Generating done
-- Build files have been written to: /app/tests/unit/build

../tests/unit $ make -C build
make: Entering directory '/app/tests/unit/build'
make[1]: Entering directory '/app/tests/unit/build'
make[2]: Entering directory '/app/tests/unit/build'
make[2]: Leaving directory '/app/tests/unit/build'
make[2]: Entering directory '/app/tests/unit/build'
[ 14%] Building C object CMakeFiles/test_uint256.dir/test_uint256.c.o
[ 28%] Building C object CMakeFiles/test_uint256.dir/app/src_common/uint256.c.o
/app/src_common/uint256.c:23:10: fatal error: read.h: No such file or directory
   23 | #include "read.h"
      |          ^~~~~~~~
compilation terminated.
make[2]: *** [CMakeFiles/test_uint256.dir/build.make:90: CMakeFiles/test_uint256.dir/app/
src_common/uint256.c.o] Error 1
make[2]: Leaving directory '/app/tests/unit/build'
make[1]: *** [CMakeFiles/Makefile2:85: CMakeFiles/test_uint256.dir/all] Error 2
make[1]: Leaving directory '/app/tests/unit/build'
make: *** [Makefile:101: all] Error 2
make: Leaving directory '/app/tests/unit/build'
```

Running `test_tx_parser` in isolation works with no issues.

Recommendation

Fix the `test_uint256` unit test.

3.3.2 Integration test

Integration tests are located in `tests/python/`. The codebase includes 10 test suites backed by snapshots using Ragger, running across all 4 devices. It also includes 20 test suites not backed by snapshots. The test results are shown in Appendix B.2.

INFO	**QB-4** Integration tests fail to run.
Perimeter	Testing

Description

2 imports in `test_eip712.py` were commented out in the source file, preventing the test suite to run. These imported names are not used, so they can be removed, letting us run the test suite.

See Appendix B.1 for the test logs.

Recommendation

Remove the deprecated imports.

```diff
--- a/tests/python/test_eip712.py
+++ b/tests/python/test_eip712.py
@@ -19,7 +19,6 @@
 import apps.eth_response_parser as ResponseParser
 from apps.eth_utils import recover_message

-from apps.celo import TrustedNameType, TrustedNameSource
 from apps.eip712 import InputData

 from apps.celo_settings import SettingID, settings_toggle
```

3.3.2.1 Snapshot-backed tests

The following snapshots are testing in some of the tests.

- **[1/1]** `celo_derive_address` (run 4 times: with and without displaying, with and without chaincode)
- **[3/3]** `eip712_advanced_filtering`
- **[2/2]** `eip712_advanced_missing_token`
- **[1/1]** `eip712_filtering_empty_array`
- **[1/1]** `sign_data`
- **[1/1]** `sign_transaction_cip64`
- **[1/1]** `sign_transaction_eip1559_no_data`

3.3.2.2 Non-snapshot-backed tests

- [1 / 1] add_cUSD_as_fee_currency
- [1 / 1] celo_get_version
- [1 / 1] eip712_bs_not_activated_error
- [38+22/60] eip712_new
- [1 / 1] eip712_skip
- [1 / 1] eip712_v0
- [0 / 1] gcs_1inch
- [1 / 1] sign_transaction_eip1559_with_data
- [1 / 1] sign_transaction_activate (legacy)
- [1 / 1] sign_transaction_create (legacy)
- [1 / 1] sign_transaction_empty (legacy)
- [1 / 1] sign_transaction_lock (legacy)
- [1 / 1] sign_transaction_no_gtw2 (legacy)
- [1 / 1] sign_transaction_no_gtw (legacy)
- [1 / 1] sign_transaction_relock (legacy)
- [1 / 1] sign_transaction_revoke (legacy)
- [1 / 1] sign_transaction_token_transfer (legacy)
- [1 / 1] sign_transaction_unlock (legacy)
- [1 / 1] sign_transaction_vote (legacy)
- [1 / 1] sign_transaction_withdraw (legacy)

INFO	QB-5 Integration test `gcs_1inch` fails.
Perimeter	Testing

Description

The application fails to parse the payload of the `gcs_1inch` test, returning `0x6a80` (`INVALID_DATA`).

```
____________________________ test_gcs_1inch[stax] ____________________________
tests/python/test_gcs.py:82: in test_gcs_1inch
    with app_client.sign("m/44'/60'/0'/0/0", tx_params, mode=SignMode.STORE):
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
---------------------------- Captured stderr call ----------------------------
[INFO] ragger.apdu_logger - => e0040001ff058000002c8000003c80000000000000000000
00002f901700181eb8502540be40085174876e80082abe194111111125421ca6dc452d289314280a
0f8842a6580b9014407ed237900000000000000000000000000f313b370d28760b98a2e935e56be92f
eb2c4ec04000000000000000000000000eeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee0000000
00000000000000000a0b86991c6218b36c1d19d4a2e9eb0ce3606eb48000000000000000000000000
0f313b370d28760b98a2e935e56be92feb2c4ec04000000000000000000000000dad77910dbdfde7
64fc21fcd4e74d71bbaca6d8d00000000000000000000000000000000000000000000000000
[INFO] ragger.apdu_logger - <= 6a80
```

We were unable to find the root cause in the alloted time frame.

Recommendation

3.4 GitHub Actions

8 GitHub Actions are configured:

- **[FAIL]** Ledger's `guidelines_enforcer.yml`;
- **[PASS]** static analysis using CodeQL (different from Ledger's version);
- **[FAIL]** build and functional tests using Ledger's reusable workflows;
- **[FAIL]** unit tests;
- **[PASS]** documentation generation (Doxygen);
- **[FAIL]** Ledger's reusable code style linter;
- **[FAIL]** Python linter (`pylint`) and type checker (`mypy`);
- **[FAIL]** spellchecking;

INFO	**QB-6** Failing GitHub Actions in CI.
Perimeter	CI

Description

6 out of 8 GitHub Actions fail.

- `guidelines_enforcer.yml` fails for 3 reasons:
  - because of the misconfigured application flags reported in Section 3.1 ,
  - because of the compiler warnings reported in Section 3.2 , and
  - because `flex_app_celo.gif` has more than 16 colors.
- `build_and_functional_tests.yml` fails at test time, because of the issue reported in Section 3.3.2 .
- `unit_test.yml` fails because of the issue reported in Section 3.3.1 .
- `coding_style_checks.yml` fails because the code is not properly linted. Details are available in the workflow logs.
- `python_client_checks.yml` fails because the Python code is neither linted nor typed. Details are available in the workflow logs.
- `misspellings_checks.yml` fails because of a lorem ipsum.

Recommendation

Fix the related findings, enforce better coding style, and remove superfluous Actions.

4 Static analysis

4.1 Clang static analyzer

No warnings/errors have been found by running the static analyzer.

4.2 CodeQL analysis

We ran CodeQL with the C++ `security-and-quality` queries. No findings have a security impact.

Name	Occurrences
Poorly documented large function	`src/handler/sign_personal_message.c:28`

Table 1: Warnings emitted by CodeQL

Name	# occurrences
Empty branch of conditional	3
Commented-out code	54
Missing header guard	1
Long switch case	1
Unused static variable	3
Total	62

Table 2: Notes emitted by CodeQL

5 Manual review

5.1 APDU overview

The host and the Personal Security Device (PSD) communicate using the Application Protocol Data Unit (APDU) protocol. This application uses `0xe0` in the `CLA` field for all commands.

When several APDU packets are required, different commands are expected to be called to populate fields in the static `tmpCtx` union variable.

src/types.h

```c
typedef union {
    publicKeyContext_t publicKeyContext;
    transactionContext_t transactionContext;
    messageSigningContext_t messageSigningContext;
    messageSigningContext712_t messageSigningContext712;
} tmpCtx_t;
```

5.2 Transaction fields displayed

Instruction	Communication	Displayed field
`INS_GET_APP_TYPE`	**STATELESS**	None.
`INS_GET_APP_CONFIGURATION`	**STATELESS**	None.
`INS_GET_PUBLIC_KEY`	**STATELESS**	None if `P1` is 0, the address otherwise.
`INS_SIGN`	**STATEFUL**	Amount, address, maximum fees.
`INS_SIGN_PERSONAL_MESSAGE`	**STATEFUL**	Message hash.
`INS_SIGN_EIP_712_MESSAGE`	**STATEFUL**	Legacy (blind signing enabled): • domain hash; • message hash. Current: • EIP-712 fields; • message hash (verbose);
`INS_PROVIDE_ERC20_TOKEN_INFORMATION`	**STATEFUL**	None.
`INS_EIP712_STRUCT_DEF`	**STATEFUL**	None.
`INS_EIP712_STRUCT_IMPL`	**STATEFUL**	Structure (verbose enabled).
`INS_EIP712_FILTERING`	**STATEFUL**	None.

5.3 Code review

Some commands interact with one-another using the `tmpCtx` union (see Section 5.1). Type confusion is prevented by using a discriminant `appState`.

5.3.1 INS_GET_APP_TYPE

This command returns the application type `0x02`.

No information is shown to the user. No user interaction is required.

INFO	**QB-7** Non-functional command `INS_GET_APP_TYPE`.
Perimeter	Command

Description

The command returns nothing instead of the intended `0x02`, because `handleGetAppType()` throws instead of returning normally.

Valid APDUs of INS_GET_APP_TYPE

```
< e0ff000000
    CLA=e0 | INS=ff | P1=00 | P2=00 | L=00 | Payload=
> 9000
    SW=9000
```

Recommendation

Remove the `THROW` instruction.

5.3.2 INS_GET_APP_CONFIGURATION

This command returns whether contract calldata is shown to the user (setting), that the application requires external token information, and the application version.

No information is shown to the user. No user interaction is required.

INFO	QB-8 Non-functional command `INS_GET_APP_CONFIGURATION`.
Perimeter	Command

Description

The command returns nothing instead of the intended `0x02010400` or `0x03010400`, because `handleGetAppConfiguration()` throws instead of returning normally.

Valid APDUs of `INS_GET_APP_TYPE`

```
< e006000000
    CLA=e0 | INS=06 | P1=00 | P2=00 | L=00 | Payload=
> 9000
    SW=9000
```

Recommendation

Remove the `THROW` instruction.

5.3.3 `INS_GET_PUBLIC_KEY`

This command returns the public key derived at the provided path. `P2` controls whether the chaincode is returned as well.

If `P1` is non-zero, the address is shown to the user for confirmation.

MEDIUM	QB-9 Cryptographic secrets may not be zeroed in memory.		
Likelihood	●○○○	**Impact**	●●●○
Perimeter	Cryptography		

Description

The `handleGetPublicKey()` function may return before erasing the populated `privateKey` and `privateKeyData` with `explicit_bzero()`.

This can happen when one of `os_derive_bip32_no_throw()`, `cx_ecfp_init_private_key_no_throw()`, or `cx_ecfp_generate_pair_no_throw` fail.

Recommendation

Erase the secrets systematically before returning from `handleGetPublicKey()`.

Valid APDUs of INS_GET_APP_TYPE (without chaincode)

```
< e00200000d038000002c8000ce1080003039
    CLA=e0 | INS=02 | P1=00 | P2=00 | L=0d | Payload=038000002c8000ce1080003039
    Path=03 8000002c 8000ce10 80003039 (m/44'/52752'/12345')
> 4104f3c5b892381bdc277026f0675634fe6b8f339a58e689faede9be2ba5701fd215599bae80
  ... 2930025fc0bd270136d9081266741d8bf562e5ca70da9ff95a1942bc2846343935466332
  ... 6243313832383145333532446266323830393934323663303432373632363744639000
    Public key size=41 (65)
    Public key=04f3c5..1942bc
    Address size=28 (40)
    Address=463439..374463 (F495Fc2bC18281E352Dbf28099426c04276267Dc)
    SW=9000
```

Valid APDUs of INS_GET_APP_TYPE (with chaincode)

```
< e00200010d038000002c8000ce1080003039
    CLA=e0 | INS=02 | P1=00 | P2=01 | L=0d | Payload=038000002c8000ce1080003039
    Path=03 8000002c 8000ce10 80003039 (m/44'/52752'/12345')
> 4104f3c5b892381bdc277026f0675634fe6b8f339a58e689faede9be2ba5701fd215599bae80
  ... 2930025fc0bd270136d9081266741d8bf562e5ca70da9ff95a1942bc2846343935466332
  ... 6243313832383145333532446266323830393934323663303432373632363744639cf6ebb
  ... 04312ba91250c9f32f2319bd3418342767459e437f30e550ffabdb2dae9000
    Public key size=41 (65)
    Public key=04f3c5..1942bc
    Address size=28 (40)
    Address=463439..374463 (F495Fc2bC18281E352Dbf28099426c04276267Dc)
    Chaincode=cf6ebb..db2dae
    SW=9000
```

5.3.4 INS_SIGN

This command parses the given transaction, shows it to the user, and returns a signature for it. It can be called several times if the transaction is too long to fit in a single APDU packet. The first packet needs to have `P1` set to `0x00`, and to contain the derivation path as well as the transaction type (EIP-1559 or CIP-64). The following packets need to have `P1` set to `0x80`.

Valid APDUs of INS_SIGN

```
< e00400007c038000002c8000003c8000000002f86c82aef380830f42408506fc35fb80825208
  ... 94da52c9ffebd4d54c94a072776126069d43e74f9e8080c080a099059ce0f1fe1f4fe27a
  ... 583a6fd6a12274780d358f332d6e5901953900b8fb22a046ce6d625369fdc8a521c22793
  ... d188afbf61500cd3095fc09b761b518560f101
    CLA=e0 | INS=04 | P1=00 | P2=01 | L=7c | Payload=03800..60f101
    Path=03 8000002c 8000003c 80000000 (m/44'/60'/0')
    TX type=02 (EIP-1559)
    RLP data=f8 6c
      82 aef3
      80
      83 0f4240
      85 06fc35fb80
      82 5208
      94 da52c9ffebd4d54c94a072776126069d43e74f9e
      80
      80
      c0
        80
        a0 99059ce0f1fe1f4fe27a583a6fd6a12274780d358f332d6e5901953900b8fb22
        a0 46ce6d625369fdc8a521c22793d188afbf61500cd3095fc09b761b518560f101
> 014bef650dcad77df8f5cb9385fca75612d3ba5abd1cae4781f1970e67a0ca19af3a4c9485d7
  ... e006c45e6907a698fd9cb1b77b162ad1f93d038985a897b32e0db99000
    Signature v=01
    Signature r=4bef65..ca19af
    Signature s=3a4c94..2e0db9
    SW=9000
```

5.3.5 INS_SIGN_PERSONAL_MESSAGE

This command signs the hash of the provided message.

HIGH	QB-10	User can sign message hashes without enabling blind signing.	
Likelihood	●●○○	**Impact**	●●●●
Perimeter	Blind signing		

Description

The command displays the hash of the message irrespective of the blind signing setting. Non-expert users can easily sign malicious messages unknowingly.

Recommendation

Disallow the command if blind signing is disabled.

Valid APDUs of `INS_SIGN_PERSONAL_MESSAGE`

```
< e00800001b038000002c8000ce10800030390000000a31323334353637383930
    CLA=e0 | INS=08 | P1=00 | P2=01 | L=1b | Payload=038000..383930
    Path=03 8000002c 8000ce10 80003039 (m/44'/52752'/12345')
    Message size=0000000a (10)
    Message=31323334353637383930 ("1234567890")
> 1c76536c41e8830a322517f921333237a616726ac151807558aeaa10f9efb8d1937299f8ad54
  ... 4f412264fb6467732bedc5215231f3dfe44444e2bc3a9597ee0eae9000
    Signature v=1c (28)
    Signature r=76536c..b8d193
    Signature s=7299f7..ee0eae
    SW=9000
```

5.3.6 `INS_SIGN_EIP_712_MESSAGE`, `INS_EIP712_*`, `INS_PROVIDE_ERC20_TOKEN_INFORMATION`

This command shows to the user the EIP-712 message stored using `INS_EIP712_STRUCT_DEF`, `INS_EIP712_STRUCT_IMPL`, `INS_EIP712_FILTERING`, and `INS_PROVIDE_ERC20_TOKEN_INFORMATION`, then returns its signature with the provided derivation path.

When `P2` is `0x00`, the legacy (v0) implementation is used. When `P2` is `0x01`, the new (full) implementation is used.

Valid APDUs of INS_SIGN_EIP_712_MESSAGE (v0)

```
... previous calls ...
< e00c00004d038000002c8000ce10800030393a79f555197402bc32e29086046c9683f812edbd
  ... d67a9a956290259c337d4c805476346eb09179b1f7d245c11a27ae6c498a419d7fad4d98
  ... 4d5b1e0ad081662a
    CLA=e0 | INS=0c | P1=00 | P2=00 | L=4d | Payload=038000..81662a
    Path=03 8000002c 8000ce10 80003039 (m/44'/52752'/12345')
    Domain hash=3a79f5..7d4c80
    Message hash=547634..81662a
> 1c78e8ef4ee101280ee1804260b73ce39cf3dc888b360f6329063fcc9df4c86a7c7a708fc717
  ... 94071c7a0026a167cc6b9dcc9e0ede28cab1add756a18317ff677d9000
    Signature v=1c (28)
    Signature r=78e8ef..c86a7c
    Signature s=7a708f..ff677d
    SW=9000
```

Valid APDUs of INS_SIGN_EIP_712_MESSAGE (full)

```
... previous calls ...
< e00c00010d038000002c8000ce1080003039
    CLA=e0 | INS=0c | P1=00 | P2=01 | L=0d | Payload=038000002c8000ce1080003039
    Path=03 8000002c 8000ce10 80003039 (m/44'/52752'/12345')
> 1c78e8ef4ee101280ee1804260b73ce39cf3dc888b360f6329063fcc9df4c86a7c7a708fc717
  ... 94071c7a0026a167cc6b9dcc9e0ede28cab1add756a18317ff677d9000
    Signature v=1c (28)
    Signature r=78e8ef..c86a7c
    Signature s=7a708f..ff677d
    SW=9000
```

6 Fuzzing

No fuzzer is present in the codebase. We hacked together a minimal fuzzer targeting `processTx()`, which immediately found 2 undefined behaviors.

```
.../src_common/rlp.c:79:43: runtime error:
 left shift of 255 by 24 places cannot be represented in type 'int'
SUMMARY: UndefinedBehaviorSanitizer: undefined-behavior .../src_common/rlp.c:79:43
.../src_common/rlp.c:54:43: runtime error:
 left shift of 145 by 24 places cannot be represented in type 'int'
SUMMARY: UndefinedBehaviorSanitizer: undefined-behavior .../src_common/rlp.c:54:43
```

MEDIUM	QB-11 Undefined behaviors in `src_common/rlp.c`.		
Likelihood	●○○○	**Impact**	●●●○
Perimeter	Parser		

Description

In `src_common/rlp.c`'s `rlpDecodeLength()`, missing type conversions lead to Undefined Behavior.

```c
*fieldLength = (*(buffer + 1) << 24) + (*(buffer + 2) << 16) +
          (*(buffer + 3) << 8) + *(buffer + 4);
```

Recommendation

Cast the read buffer to a (large enough) unsigned type before applying the shift.

```c
*fieldLength = ((uint32_t)*(buffer + 1) << 24) + (*(buffer + 2) << 16) +
          (*(buffer + 3) << 8) + *(buffer + 4);
```

7 Conclusion

7.1 Issues summary

ID	Name	Perimeter
HIGH-10	User can sign message hashes without enabling blind signing.	Blind signing
MEDIUM-9	Cryptographic secrets may not be zeroed in memory.	Cryptography
MEDIUM-11	Undefined behaviors in `src_common/rlp.c`.	Parser
INFO-1	Outdated flag in LedgerHQ's application database.	Configuration
INFO-2	C compiler raises 1 warning.	Compilation
INFO-3	Unit test `test_uint256` fails to compile.	Compilation, Test
INFO-4	Integration tests fail to run.	Testing
INFO-5	Integration test `gcs_1inch` fails.	Testing
INFO-6	Failing GitHub Actions in CI.	CI
INFO-7	Non-functional command `INS_GET_APP_TYPE`.	Command
INFO-8	Non-functional command `INS_GET_APP_CONFIGURATION`.	Command

7.2 Recommendations

ID	Name
HIGH-10	Disallow the command if blind signing is disabled.
MEDIUM-9	Erase the secrets systematically before returning from `handleGetPublicKey()`.
MEDIUM-11	Cast the read buffer to a (large enough) unsigned type before applying the shift. `*fieldLength = ((uint32_t)*(buffer + 1) << 24) + (*(buffer + 2) << 16) + (*(buffer + 3) << 8) + *(buffer + 4);` (C)
INFO-1	Ledger should update the database entry corresponding to CELO.
INFO-2	Remove the unused variable and unused parameters, or mark them with `__attribute__((unused))` if this is intended.
INFO-3	Fix the `test_uint256` unit test.
INFO-4	Remove the deprecated imports. `--- a/tests/python/test_eip712.py` (diff)

	<pre>+++ b/tests/python/test_eip712.py
	@@ -19,7 +19,6 @@
	 import apps.eth_response_parser as ResponseParser
	 from apps.eth_utils import recover_message
	
	-from apps.celo import TrustedNameType, TrustedNameSource
	 from apps.eip712 import InputData
	
	 from apps.celo_settings import SettingID, settings_toggle</pre>

INFO-5	
INFO-6	Fix the related findings, enforce better coding style, and remove superfluous Actions.
INFO-7	Remove the `THROW` instruction.
INFO-8	Remove the `THROW` instruction.

8 Ledger summary

8.1 Purpose of the update

New application (not an update).

8.2 Issue summary

ID	Name	Perimeter
HIGH-10	User can sign message hashes without enabling blind signing.	Blind signing
MEDIUM-9	Cryptographic secrets may not be zeroed in memory.	Cryptography
MEDIUM-11	Undefined behaviors in `src_common/rlp.c`.	Parser
INFO-1	Outdated flag in LedgerHQ's application database.	Configuration
INFO-2	C compiler raises 1 warning.	Compilation
INFO-3	Unit test `test_uint256` fails to compile.	Compilation, Test
INFO-4	Integration tests fail to run.	Testing
INFO-5	Integration test `gcs_1inch` fails.	Testing
INFO-6	Failing GitHub Actions in CI.	CI
INFO-7	Non-functional command `INS_GET_APP_TYPE`.	Command
INFO-8	Non-functional command `INS_GET_APP_CONFIGURATION`.	Command

8.3 Version information

Initial audit (1.0)	
Repository	https://github.com/blooo-io/app-celo-spender.git
Version	`N/A`
Commit	`55fdbcaad1336d121a3cf8895c0d3ee5ba933a1f`

8.4 Ledger DevPortal checklist

- [ ] Application flags, derivation paths and curves are correctly set.
- [x] Clear signing correctly displays information and asks for user confirmation.
- [ ] Blind signing is disabled by default or does not exist (if it exists, it does apply for complex transactions but not to coin transfers or swaps).
- [ ] App workflow is considered secure and snapshots cover the nominal and error cases of app features.
- [x] Boilerplate Makefiles and SDK are used to compile the app. If it's a new app, it has been developed over the C/Rust boilerplates.
- [ ] The app compiles without errors or warnings (no errors/warnings are silenced using compilation flags or pragma directives).
- [ ] Guidelines enforcer and build workflows are integrated and passed.
- [ ] Unit tests, functional tests and fuzzing tests are passing.
- [ ] ~~For app-plugins, the generic fuzzing target is integrated~~. This is not a plugin.
- [x] scan-build/CodeQL or clippy succeed without returning vulnerabilities. → *Reported elements are false positives, non-relevant, or from non-default "restriction" lints.*
- [x] Dependencies are needed, maintained and secure (no known impactful vulnerability for the used version).
- [ ] The app does not contain any medium or high severity vulnerabilities.
- [x] The review process looked into technical vulnerabilities as well as logical ones and weaknesses.
- [ ] Secrets are correctly erased from memory, not exported or shown to the user.
- [x] User supplied messages use a prefix and are validated before signature.
- [x] No cryptographic primitives are re-implemented if they already exist in BOLOS or the SDK.
- [x] The client is not able to freely manipulate the device keypairs.
- [x] An authenticated encryption scheme is used for data being cached on the client.

8.5 Production Ready

The application is **not production ready**.

8.6 Remarks

The audited version is an older version than the tip of the repository due to a bug in the most recent version.

Appendix

A Build configuration

Makefile

```
CURVE_APP_LOAD_PARAMS = secp256k1
…                                                    …
PATH_APP_LOAD_PARAMS = "44'/52752'" "44'/60'/0'/0/0" "44'/60'/0'" "44'/60'/0'/0"
…                                                    …
VARIANT_PARAM = CHAIN
VARIANT_VALUES = celo
…                                                    …
ENABLE_BLUETOOTH = 1
#ENABLE_NFC = 1
ENABLE_NBGL_FOR_NANO_DEVICES = 1
…                                                    …
ENABLE_NBGL_QRCODE = 1
#ENABLE_NBGL_KEYBOARD = 1
#ENABLE_NBGL_KEYPAD = 1
```

ledger_app.toml

```
[app]
build_directory = "./"
sdk = "C"
devices = ["nanox", "flex","nanos+", "stax"]

[tests]
unit_directory = "./tests/unit"
pytest_directory = "./tests/python"
```

B Integration test logs

B.1 Failing to run

```
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /srv/ub-venv/bin/python3
cachedir: .pytest_cache
rootdir: /srv
configfile: pytest.ini
plugins: cov-7.0.0

collecting ... collecting 6 items

collected 23 items / 1 error

=================================== ERRORS ====================================
_________________ ERROR collecting tests/python/test_eip712.py _________________
ImportError while importing test module '/srv/tests/python/test_eip712.py'.
Hint: make sure your test modules/packages have valid Python names.
Traceback:
/usr/lib/python3.12/importlib/__init__.py:90: in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
tests/python/test_eip712.py:22: in <module>
    from apps.celo import TrustedNameType, TrustedNameSource
E   ImportError: cannot import name 'TrustedNameType' from 'apps.celo' (/srv/tests/python/apps/celo.py)
=============================== warnings summary ===============================
ub-venv/lib/python3.12/site-packages/websockets/legacy/__init__.py:6
  /srv/ub-venv/lib/python3.12/site-packages/websockets/legacy/__init__.py:6: DeprecationWarning: websockets.legacy
is deprecated; see https://websockets.readthedocs.io/en/stable/howto/upgrade.html for upgrade instructions
    warnings.warn(  # deprecated in 14.0 - 2024-11-09

ub-venv/lib/python3.12/site-packages/ragger/firmware/structs.py:36
ub-venv/lib/python3.12/site-packages/ragger/firmware/structs.py:36
  /srv/ub-venv/lib/python3.12/site-packages/ragger/firmware/structs.py:36: UserWarning: `ragger.firmware.Firmware`
is deprecated, use `ledgered.devices.Devices` or `ledgered.devices.DeviceType` instead
    warn(DEPRECATION_MESSAGE)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================== short test summary info ============================
ERROR tests/python/test_eip712.py
!!!!!!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!
======================== 3 warnings, 1 error in 1.22s =========================
```

B.2 Results

```
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /app/venv/bin/python3
cachedir: .pytest_cache
rootdir: /app
configfile: pytest.ini
plugins: cov-7.0.0

collecting ... collecting 6 items

collected 92 items

tests/python/test_celo.py::test_celo_derive_address[stax-stax-False-False] PASSED [  1%]
tests/python/test_celo.py::test_celo_get_version[stax-stax] PASSED       [  2%]
tests/python/test_celo.py::test_sign_data[stax-stax] PASSED              [  3%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_empty[stax-stax] PASSED [  4%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_token_transfer[stax-stax] PASSED [  5%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_lock[stax-stax] PASSED [  6%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_vote[stax-stax] PASSED [  7%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_activate[stax-stax] PASSED [  8%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_revoke[stax-stax] PASSED [  9%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_unlock[stax-stax] PASSED [ 10%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_withdraw[stax-stax] PASSED [ 11%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_relock[stax-stax] PASSED [ 13%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_create[stax-stax] PASSED [ 14%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_no_gtw[stax-stax] PASSED [ 15%]
tests/python/test_deprecated_legacy.py::test_sign_transaction_no_gtw2[stax-stax] PASSED [ 16%]
tests/python/test_sign_new_transaction.py::test_sign_transaction_eip1559_no_data[stax-stax] PASSED [ 17%]
tests/python/test_sign_new_transaction.py::test_sign_transaction_cip64[stax-stax] PASSED [ 18%]
tests/python/test_eip712.py::test_eip712_v0[stax] PASSED                 [ 19%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/00-simple_mail-data.json-True-False-stax] PASSED
 [ 20%]
tests/python/test_eip712.py::test_eip712_advanced_filtering[data_set0-stax] PASSED [ 21%]
tests/python/test_eip712.py::test_eip712_filtering_empty_array[stax] PASSED [ 22%]
tests/python/test_eip712.py::test_eip712_advanced_missing_token[tokens0-stax] PASSED [ 23%]
tests/python/test_eip712.py::test_eip712_bs_not_activated_error[stax] PASSED [ 25%]
tests/python/test_eip712.py::test_eip712_skip[stax] PASSED               [ 26%]
tests/python/test_gcs.py::test_gcs_1inch[stax] FAILED                    [ 27%]
tests/python/test_sign_new_transaction.py::test_sign_transaction_eip1559_with_data[stax] PASSED [ 28%]
tests/python/test_sign_new_transaction.py::test_add_cUSD_as_fee_currency[stax] PASSED [ 29%]
tests/python/test_celo.py::test_celo_derive_address[stax-stax-False-True] PASSED [ 30%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/00-simple_mail-data.json-True-True-stax] PASSED
 [ 31%]
tests/python/test_eip712.py::test_eip712_advanced_filtering[data_set1-stax] PASSED [ 32%]
tests/python/test_eip712.py::test_eip712_advanced_missing_token[tokens1-stax] PASSED [ 33%]
tests/python/test_celo.py::test_celo_derive_address[stax-stax-True-False] PASSED [ 34%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/00-simple_mail-data.json-False-False-stax] PASSED
 [ 35%]
tests/python/test_eip712.py::test_eip712_advanced_filtering[data_set2-stax] PASSED [ 36%]
tests/python/test_celo.py::test_celo_derive_address[stax-stax-True-True] PASSED [ 38%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/00-simple_mail-data.json-False-True-stax] PASSED
 [ 39%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/01-addresses_array_mail-data.json-True-False-
stax]
PASSED [ 40%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/01-addresses_array_mail-data.json-True-True-stax]
SKIPPED [ 41%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/01-addresses_array_mail-data.json-False-False-
stax]
PASSED [ 42%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/01-addresses_array_mail-data.json-False-True-
stax]
SKIPPED [ 43%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/02-recipients_array_mail-data.json-True-False-
stax]
PASSED [ 44%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/02-recipients_array_mail-data.json-True-True-
stax]
SKIPPED [ 45%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/02-recipients_array_mail-data.json-False-False-
stax]
PASSED [ 46%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/02-recipients_array_mail-data.json-False-True-
stax]
SKIPPED [ 47%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/03-long_string-data.json-True-False-stax] PASSED
 [ 48%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/03-long_string-data.json-True-True-stax] SKIPPED
 [ 50%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/03-long_string-data.json-False-False-stax] PASSED
 [ 51%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/03-long_string-data.json-False-True-stax] SKIPPED
 [ 52%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/04-long_bytes-data.json-True-False-stax] PASSED
 [ 53%]
```

```
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/04-long_bytes-data.json-True-True-stax] SKIPPED
 [ 54%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/04-long_bytes-data.json-False-False-stax] PASSED
 [ 55%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/04-long_bytes-data.json-False-True-stax] SKIPPED
 [ 56%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/05-signed_ints-data.json-True-False-stax] PASSED
 [ 57%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/05-signed_ints-data.json-True-True-stax] SKIPPED
 [ 58%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/05-signed_ints-data.json-False-False-stax] PASSED
 [ 59%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/05-signed_ints-data.json-False-True-stax] SKIPPED
 [ 60%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/06-boolean-data.json-True-False-stax] PASSED
 [ 61%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/06-boolean-data.json-True-True-stax] SKIPPED
 [ 63%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/06-boolean-data.json-False-False-stax] PASSED
 [ 64%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/06-boolean-data.json-False-True-stax] SKIPPED
 [ 65%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/07-fixed_bytes-data.json-True-False-stax] PASSED
 [ 66%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/07-fixed_bytes-data.json-True-True-stax] SKIPPED
 [ 67%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/07-fixed_bytes-data.json-False-False-stax] PASSED
 [ 68%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/07-fixed_bytes-data.json-False-True-stax] SKIPPED
 [ 69%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/08-opensea-data.json-True-False-stax] PASSED
 [ 70%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/08-opensea-data.json-True-True-stax] PASSED [ 71%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/08-opensea-data.json-False-False-stax] PASSED
 [ 72%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/08-opensea-data.json-False-True-stax] PASSED
 [ 73%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/09-rarible-data.json-True-False-stax] PASSED
 [ 75%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/09-rarible-data.json-True-True-stax] SKIPPED
 [ 76%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/09-rarible-data.json-False-False-stax] PASSED
 [ 77%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/09-rarible-data.json-False-True-stax] SKIPPED
 [ 78%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/10-multidimensional_arrays-data.json-True-False-
stax]
PASSED [ 79%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/10-multidimensional_arrays-data.json-True-True-
stax]
SKIPPED [ 80%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/10-multidimensional_arrays-data.json-False-False-
stax]
PASSED [ 81%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/10-multidimensional_arrays-data.json-False-True-
stax]
SKIPPED [ 82%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/11-complex_structs-data.json-True-False-stax]
PASSED [ 83%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/11-complex_structs-data.json-True-True-stax]
PASSED [ 84%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/11-complex_structs-data.json-False-False-stax]
PASSED [ 85%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/11-complex_structs-data.json-False-True-stax]
PASSED [ 86%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/12-sign_in-data.json-True-False-stax] PASSED
 [ 88%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/12-sign_in-data.json-True-True-stax] PASSED [ 89%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/12-sign_in-data.json-False-False-stax] PASSED
 [ 90%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/12-sign_in-data.json-False-True-stax] PASSED
 [ 91%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/13-empty_arrays-data.json-True-False-stax] PASSED
 [ 92%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/13-empty_arrays-data.json-True-True-stax] SKIPPED
 [ 93%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/13-empty_arrays-data.json-False-False-stax] PASSED
 [ 94%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/13-empty_arrays-data.json-False-True-stax] SKIPPED
 [ 95%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/14-rabby_bug-data.json-True-False-stax] PASSED
 [ 96%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/14-rabby_bug-data.json-True-True-stax] SKIPPED
 [ 97%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/14-rabby_bug-data.json-False-False-stax] PASSED
```

```
 [ 98%]
tests/python/test_eip712.py::test_eip712_new[/app/tests/python/eip712_input_files/14-rabby_bug-data.json-False-True-stax] SKIPPED
 [100%]

=================================== FAILURES ===================================
____________________________ test_gcs_1inch[stax] ____________________________
tests/python/test_gcs.py:82: in test_gcs_1inch
    with app_client.sign("m/44'/60'/0'/0/0", tx_params, mode=SignMode.STORE):
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
tests/python/apps/celo.py:444: in sign
    self._exchange(chunk)
tests/python/apps/celo.py:216: in _exchange
    return self._backend.exchange_raw(payload)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
venv/lib/python3.12/site-packages/ragger/backend/speculos.py:66: in decoration
    raise ExceptionRAPDU(rapdu.status, rapdu.data)
E   ragger.error.ExceptionRAPDU: Error [0x6a80] b''
--------------------------- Captured stdout setup ---------------------------

--------------------------- Captured stderr setup ---------------------------
[INFO] ragger.logger - Speculos binary: '/app/build/stax/bin/app.elf'
[INFO] ragger.logger - SDK Library: 'nbgl'
[INFO] ragger.logger - Speculos options: '--model stax --api-port 5000 --apdu-port 5001'
[INFO] ragger.logger - Starting SpeculosBackend stream
root: Parsing binary '/app/build/stax/bin/app.elf'
speculos: Api level detected from metadata: 24
root: Parsing binary '/app/build/stax/bin/app.elf'
speculos: Found C_nbgl_fonts in sharedlib stax-api-level-shared-24.elf at 0x81539C (24 bytes)

root: Parsing binary '/app/build/stax/bin/app.elf'
apdu: Server started on 0.0.0.0:5001
[*] speculos launcher revision:
[*] using API_LEVEL version 24 on stax
[*] loading CXLIB from "/app/venv/lib/python3.12/site-packages/speculos/sharedlib/stax-api-level-shared-24.elf"
[*] patching svc instruction at 0x814adc
[*] patching svc instruction at 0x80f0e4
[*] Seed initialized from environment
[*] Env app name: 'Celo'
[*] Env app version: '1.4.0'
----------------------------- Captured log setup -----------------------------
INFO     ragger.logger:speculos.py:116 Speculos binary: '/app/build/stax/bin/app.elf'
INFO     ragger.logger:speculos.py:117 SDK Library: 'nbgl'
INFO     ragger.logger:speculos.py:118 Speculos options: '--model stax --api-port 5000 --apdu-port 5001'
INFO     ragger.logger:speculos.py:169 Starting SpeculosBackend stream
INFO
 speculos-client:client.py:219 starting speculos with command: /app/venv/bin/python3 -m speculos --model stax --api-port 5000 --apdu-
port 5001 --display headless /app/build/stax/bin/app.elf
---------------------------- Captured stderr call ----------------------------
[INFO] ragger.apdu_logger - => e0040001ff058000002c8000003c8000000000000000000000002f901700181eb8502540be40085174876e80082abe19411111125421ca6dc452d289314
[INFO] ragger.apdu_logger - <= 6a80
----------------------------- Captured log call ------------------------------
INFO
 ragger.apdu_logger:speculos.py:206 => e0040001ff058000002c8000003c8000000000000000000000002f901700181eb8502540be40085174876e80082abe19411111125421ca6dc45
INFO     ragger.apdu_logger:speculos.py:63 <= 6a80
---------------------------- Captured log teardown ----------------------------
INFO     speculos-client:client.py:224 stopping speculos
============================== warnings summary ===============================
venv/lib/python3.12/site-packages/websockets/legacy/__init__.py:6
  /app/venv/lib/python3.12/site-packages/websockets/legacy/__init__.py:6: DeprecationWarning: websockets.legacy is deprecated; see
https://websockets.readthedocs.io/en/stable/howto/upgrade.html for upgrade instructions
    warnings.warn(  # deprecated in 14.0 - 2024-11-09

venv/lib/python3.12/site-packages/ragger/firmware/structs.py:36: 1 warning
tests/python/test_deprecated_legacy.py: 12 warnings
tests/python/test_sign_new_transaction.py: 2 warnings
  /app/venv/lib/python3.12/site-packages/ragger/firmware/structs.py:36: UserWarning: `ragger.firmware.Firmware` is deprecated, use
`ledgered.devices.Devices` or `ledgered.devices.DeviceType` instead
    warn(DEPRECATION_MESSAGE)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================== short test summary info ============================
FAILED tests/python/test_gcs.py::test_gcs_1inch[stax] - ragger.error.ExceptionRAPDU: Error [0x6a80] b''
====== 1 failed, 69 passed, 22 skipped, 16 warnings in 534.84s (0:08:54) =======
```